DaimlerChrysler

Adaption Module as Vehicle Control
ADM2

Control Unit **Operating Manual**

Abbreviation	Meaning
ABS	Antiblock system
ADM2	Adaption module as vehicle control, new version
ADR	PTO speed control
BK	Engine retarder flap, also MBR-BK
C3/B7	Speed signal C3/B7
CAN	Control Area Network
CC+	Cruise Control Resume and Acceleration
CC-	Cruise Control Set and Decelerate
CC_EIN	Cruise Control, Cruise control on-off switch
EEPROM	Electrical erasable and programmable read only memory
EMV	EMC/electromagnetic compatibility
EWG	European economic communities in the European Community, precursor of the EU
FFG	The foot throttle actuator is the accelerator pedal
FMR	Vehicle control for Mercedes-Benz commercial vehciles type Actros or type Atego
FSBE	Input for the switching state of the parking brake
HFG	Remote pedal
Highside Schalter	Switch (switched to battery voltage)
IWA	Actual value output
K-Leitung	Serial communication- and diagnosis line
KD	Constantly open valve, also MBR-KD
Lowside Schalter	Switch (switched to ground)
MBR	Engine brake
MCAN	Engine CAN data bus between ADM2 and PLD-MR
Minidiag 2	Diagnosis- and configuration unit for the ADM2

Abbreviation	Meaning
NE	Input for transmission position Neutral
PLD-MR	Engine control type PLD (for the injection principle pump-line-nozzle)
PTO	Power Take Off
PWM	Pulse width modulation
SAE J1939	CAN data bus according to standard SAE J1939

1. Safety

1.1. Symbols

The instructions which follow are shown against various symbols.

Risk of injury!

This symbol appears against all safety instructions which must be complied with in order to avoid a direct risk of danger to life and limb.

This symbol is used against all safety instructions which, if disregarded, could give rise to the danger of material damage or malfunctions.

1.2. General information

Risk of potentially fatal accident!

The ADM2 vehicle control adaption module is essential for defining the functions of the engine and vehicle. Functions such as engine start, engine stop, accelerator pedal evaluation, actuation of engine brake etc. are relevant to safety.
Incorrectly performed modifications to the parameters or tampering with the wiring can cause far-reaching changes to the performance of the engine and/or vehicle. This can lead to personal injury and material damage.

The ADM2 control unit has been developed and tested in accordance with the DaimlerChrysler Specifications for Operating Safety and EMC Compatibility. The manufacturer of the vehicle or equipment is solely responsible for the examination and implementation of applicable legal stipulations.

1.3. Use for the intended purpose

The DaimlerChrysler engine and the ADM2 control unit are only to be used for the purpose stated in the contract of purchase. Any other use or an extension of the stated use will be regarded as not conforming to the engine´s intended purpose.

DaimlerChrysler AG cannot accept any liability for damage resulting from such use.

Liability for damage resulting from the engine not having been used for its intended purpose shall rest solely with the manufacturer of the complete machine or vehicle in which the engine is installed.

These ADM2 Operating Instructions and the engine Operating Instructions must be observed.

1.4. Personnel requirements

Work on the electrics and programmed parameters must only be carried out by specially skilled persons or those who have received training from DaimlerChrysler, or by specialists employed by a workshop authorised by DaimlerChrysler.

1.5. Conversions and modifications to the ADM2

Unauthorised modifications to the ADM2 could affect the operation and safety of the vehicle/machine in which it is installed. No responsibility will be accepted for any resulting damage.

1.6. Installation

The guidelines and instructions in chapter 5 must be observed.

1.7. Organisational measures

These Operating Instructions should be handed to personnel entrusted with the operation of the ADM2 and should, whenever possible, be stored in an easily accessible place.

With the aid of these Operating Instructions, personnel must be familiarised with the operation of the ADM2, paying special attention to the safety-relevant instructions applicable to the engine.

This applies in particular to personnel who only work on the engine and ADM2 occasionally. In addition to these Operating Instructions, comply with local legal stipulations and any other obligatory accident prevention and environmental protection regulations which may apply in the country of operation.

1.8. Safety precautions for engines with electronic control units

Risk of accident!

When the vehicle electrics are first operated, the drive train must be open (transmission in neutral). The engine could start unexpectedly due to incorrect wiring or unsuitable parameter programming. If the drive train is closed (transmission not in neutral), the vehicle could unexpectedly start moving or set the working machine in operation, constituting a risk to life and limb.

The safety precautions stated below must be applied at all times in order to avoid damage to the engine, its components and wiring, and to avoid possible personal injury.

- Only start the engine with the batteries securely connected.
- Do not disconnect the batteries when the engine is running.
- Only start the engine with the engine speed sensor connected.
- Do not start the engine with the aid of a rapid battery charger. If emergency starting is necessary, only start using separate batteries.
- The battery terminal clamps must be disconnected before a rapid charger is used. Comply with the operating instructions for the rapid charger.
- If electric welding work is to be performed, the batteries must be disconnected and both cables (+ and -) secured together.
- Work is only to be performed on the wiring and connectors are only to be plugged/unplugged with the electrical system switched off.
- The first time starting up the engine, the possibility must be provided to switch off the voltage supply to the MR engine control and to the ADM2 adaption module in an emergency. If it is incorrectly wired up, it may no longer be possible to switch off the engine.
- Interchanging the poles of the control unit´s voltage supply (e.g. by interchanging the battery poles) can damage the control unit beyond repair.
- Fasten connectors on the fuel injection system with the specified tightening torque.
- Only use properly fitting test leads for measurements on plug connectors (DaimlerChrysler connector set).

If temperatures in excess of 80 °C (e.g. in a drying kiln) are to be expected, the control untis must be removed as they could be damaged by such temperatures.

Telephones and two-way radios which are not connected to an external aerial can cause malfunctions in the vehicle electronics and thus jeapardise the engine´s operating safety.